Monocular 3D reconstruction

Achievements

July 2024

Alberto Rota

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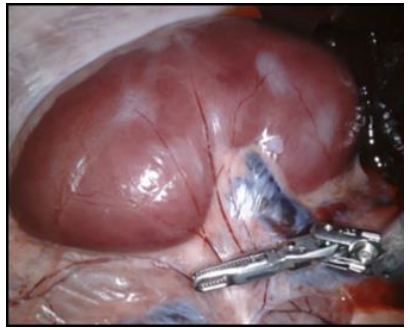
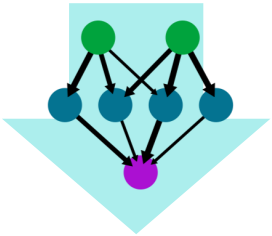
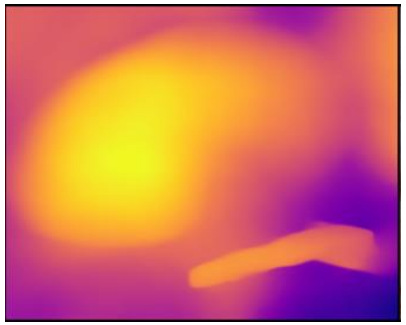
Major Achievements

Monocular Depth Estimation Network



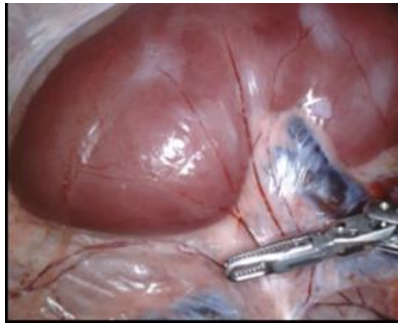
3 x 448 x 448

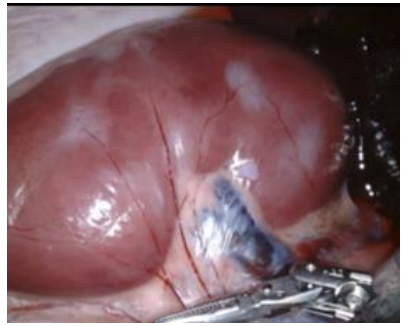




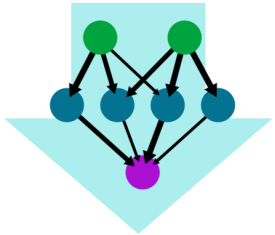
1 x 448 x 448

Monocular Visual Odometry Network

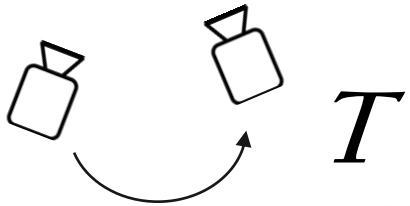




2 x 3 x 448 x 448



1 x 6



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Datasets

Training : Public datasets

- SCARED
- StereoMIS

→ Annotated with Camera Pose → From Kinematic Chain

Evaluation : Mix of public and internal dataset

- SCARED Test Split [Pose GT, NO Depth GT]
- StereoMIS Test Split [Pose GT, NO Depth GT]
- SERV-CT [NO Pose GT, Depth GT]
- Internal Asensus Dataset [Pose GT, Depth GT from stereo]

→ Comparison with Internal Asensus Stereo Reconstruction and/or State of the Art from scientific literature

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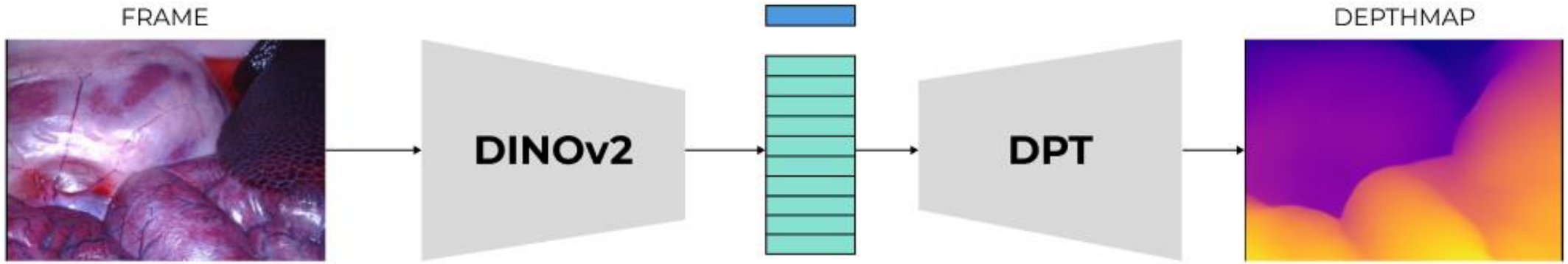
Depth Estimation Network

Architecture



Trained **self-supervised** with Odometry GT from SCARED and StereoMIS

Technicalities

- Parameters: **343M** total (303M pretrained)
- Forward pass Time: **22.91 ± 1.4 ms** / **43 fps** on NVIDIA A100 312 TFLOPS
  → Estimated 11 fps on RTX 4090 [83 TFLOPS]

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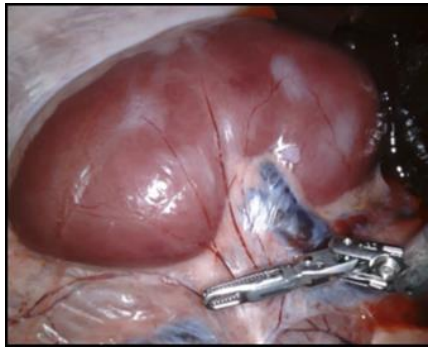
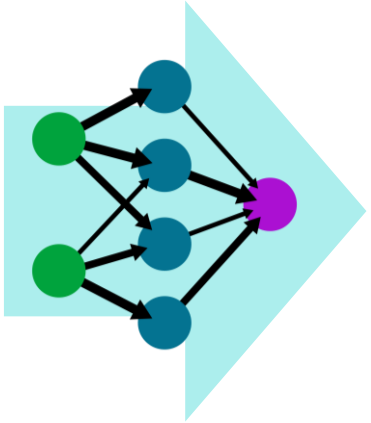
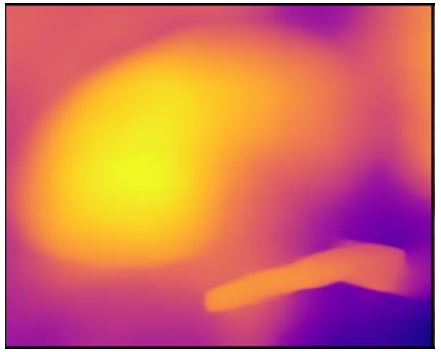
Self-Supervised Depth Estimation

From the previous frame

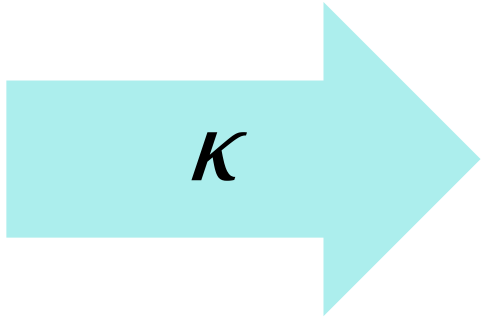




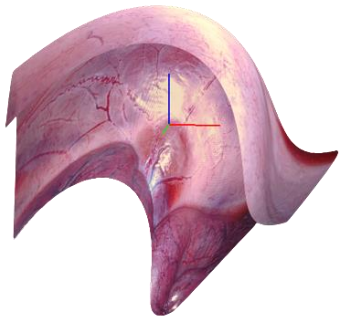
Predict the depthmap



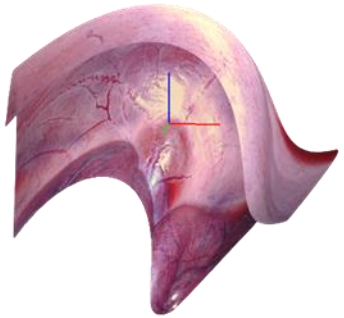
And with the Intrinsic K



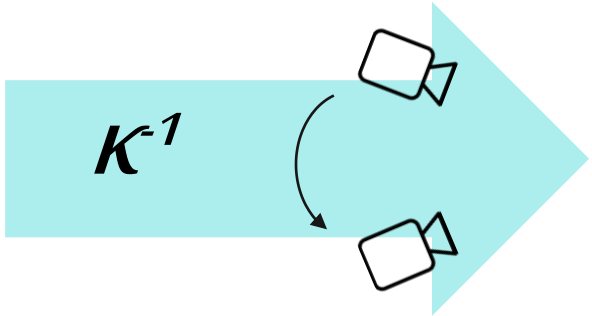
Get a RGB 3D point cloud



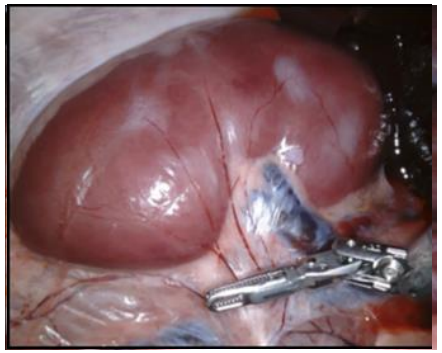
From the point cloud



The Inverse Intrinsics and the camera motion



Synthetize the current frame



The only source of error is the one made estimating the depthmap

Use this error as a loss function to Train the model

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Qualitative Inspection of DepthMaps

Cholecistectomy – Esslingen 02 Aug 2023

40 minutes procedure – Showing 1 Depthmap every 120sec

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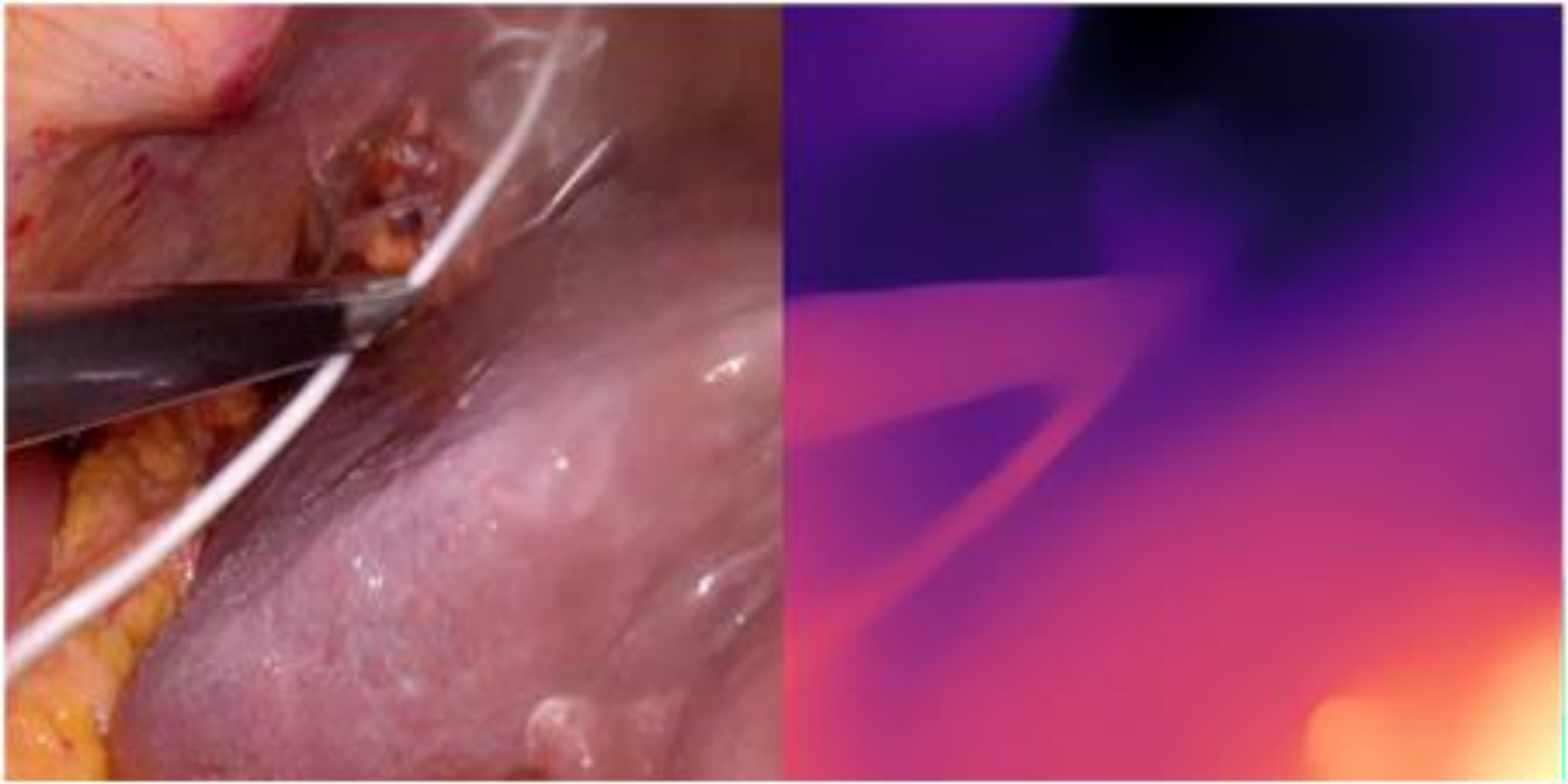

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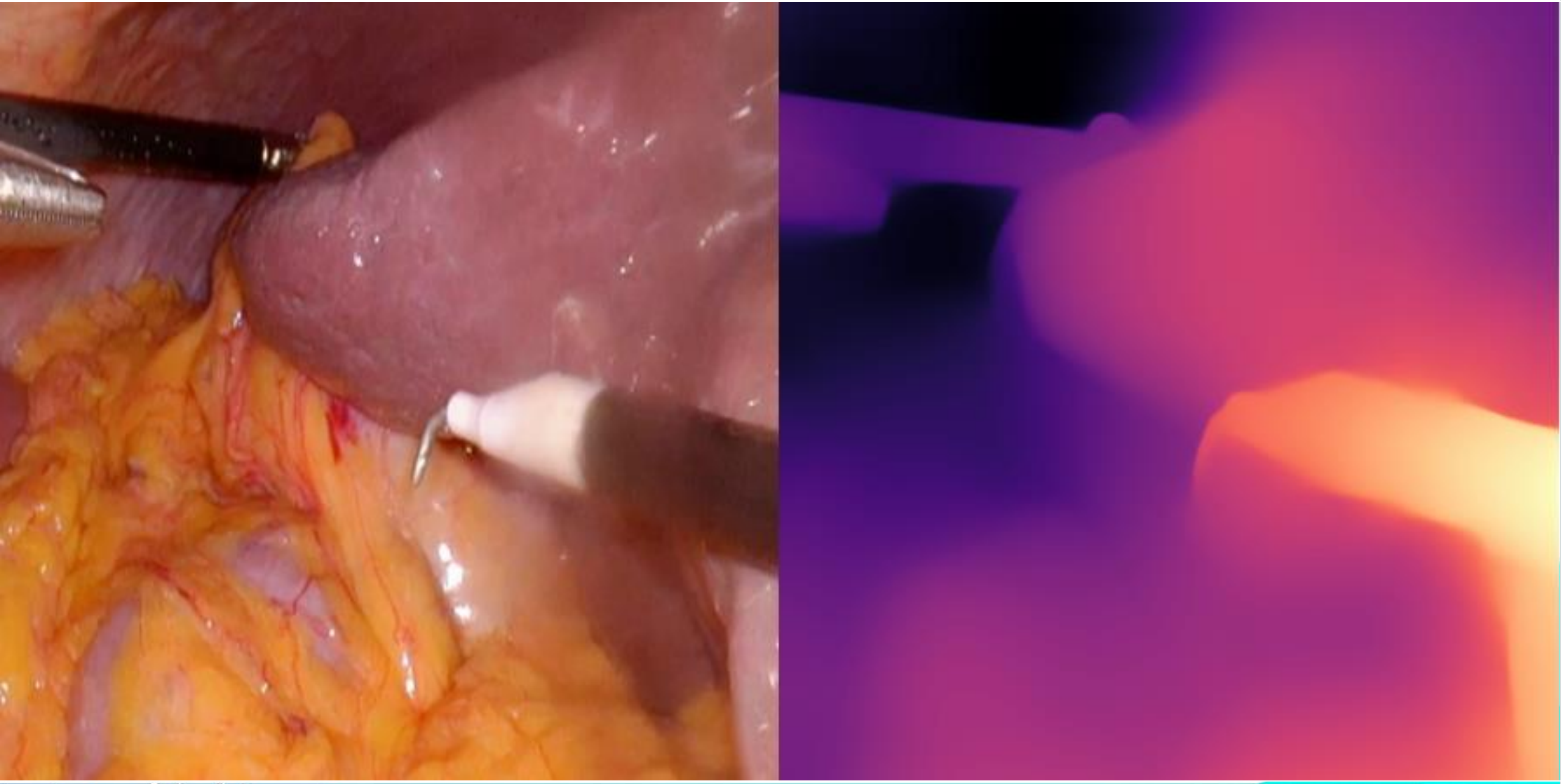

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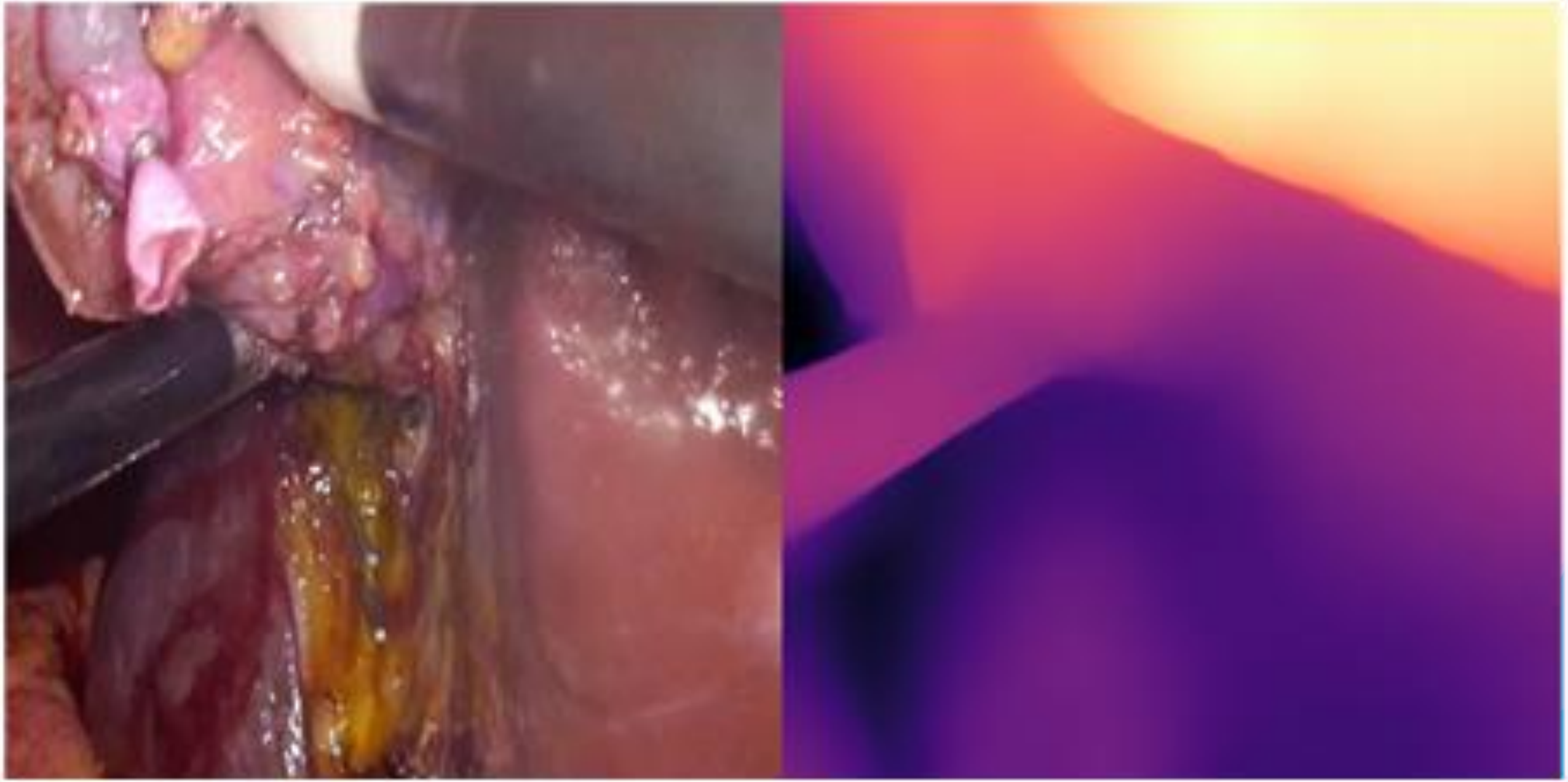

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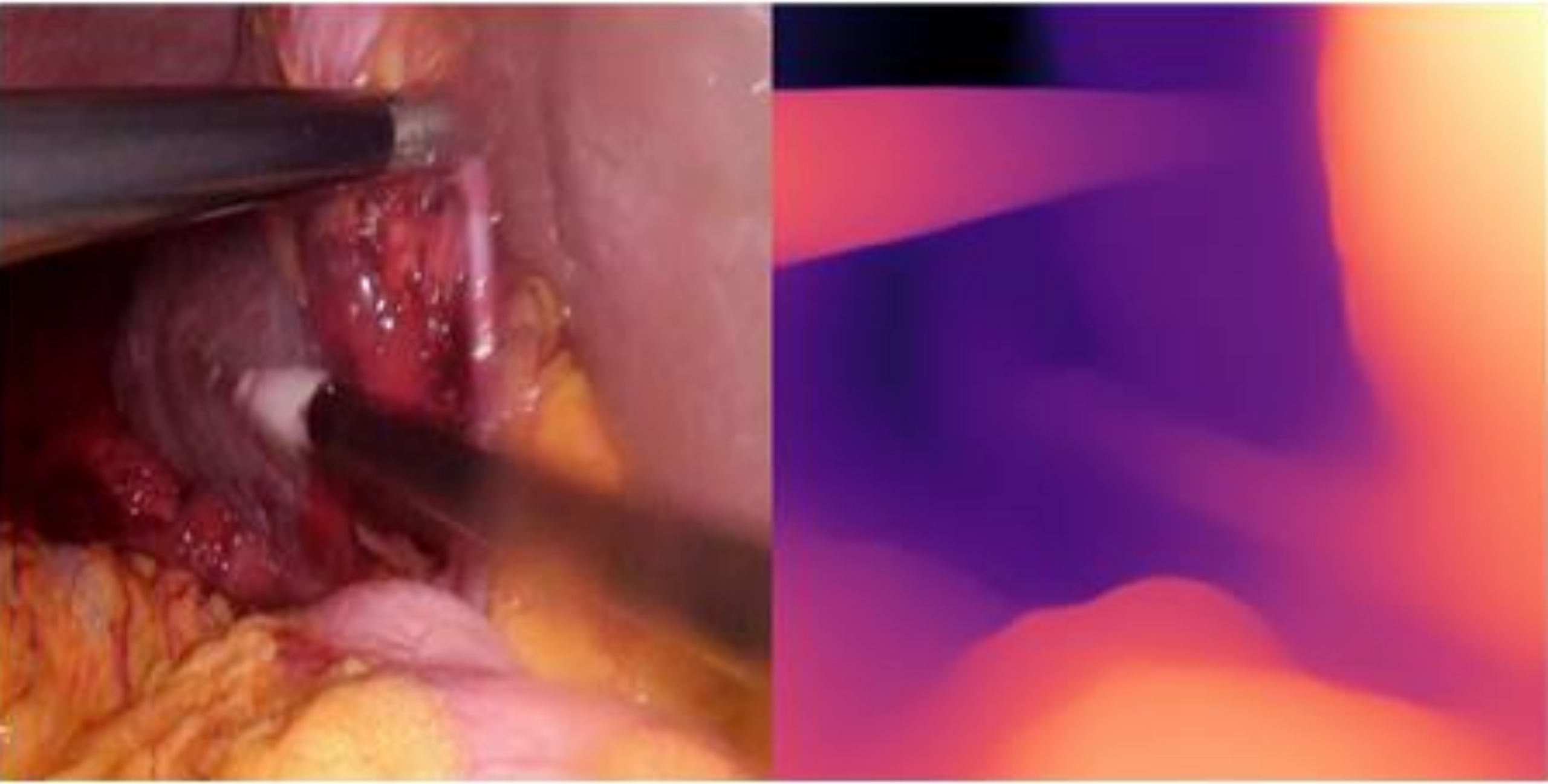

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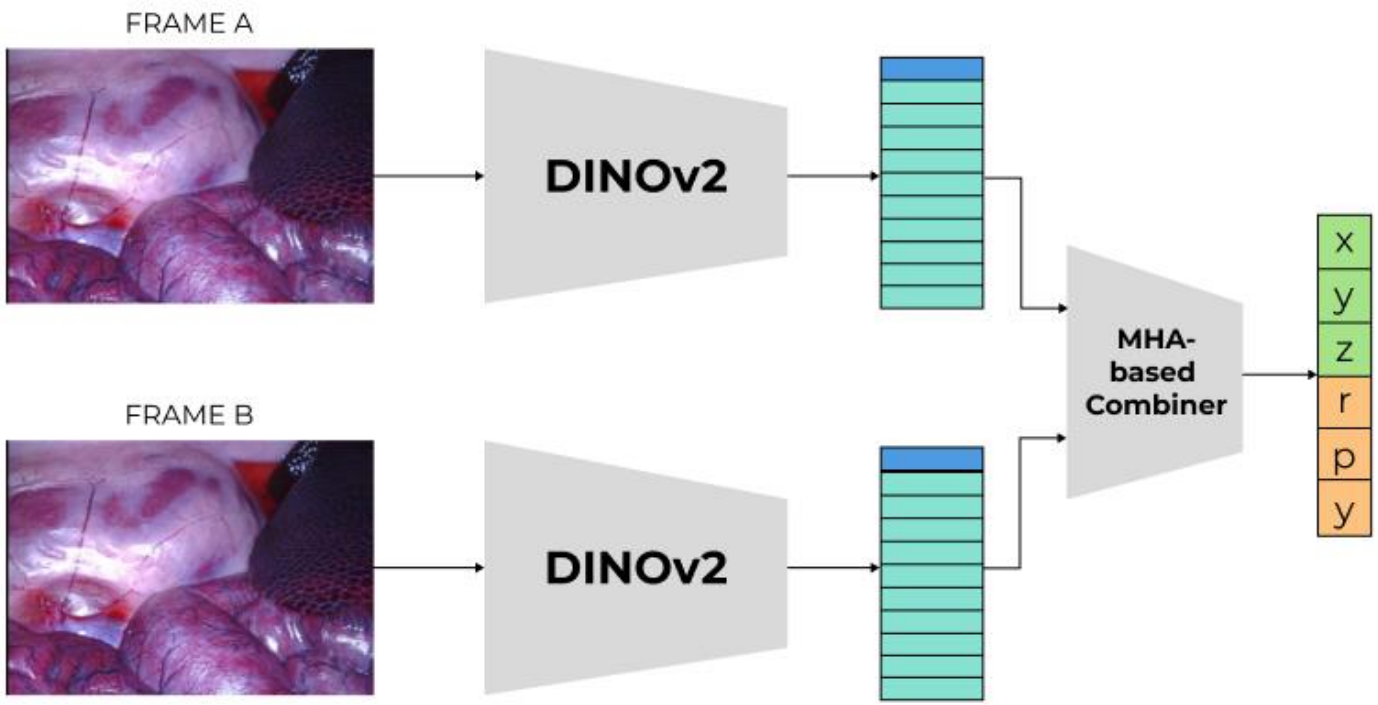
Pose Estimation

Architecture

Trained **fully supervised** with GT from SCARED and StereoMIS



Technicalities

- Parameters: **305M** total (303M pretrained)
- Forward-pass Time: **17.64 ± 2.3 ms** / **57 fps** on NVIDIA A100 312 TFLOPS]

  → Estimated 15 fps on RTX 4090 [83 TFLOPS]

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Quantitative Inspection of Trajectories

Video1 and Video2 of SCARED Test Split

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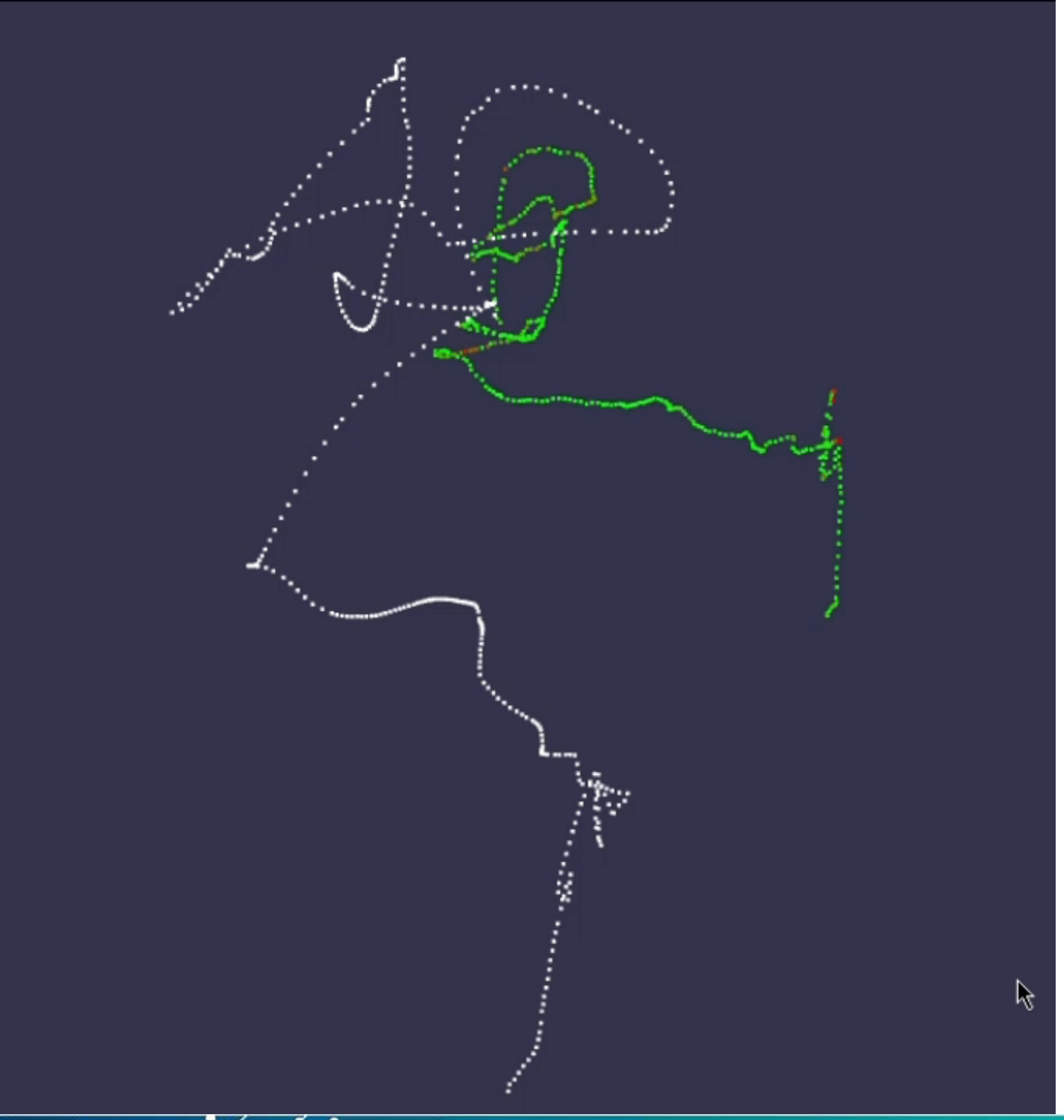

Compared to GT

- Absolute Trajectory Error: **38.62 mm**
- Relative Position Error: **1.05 mm**
- Relative Rotation Error: **3.0°**

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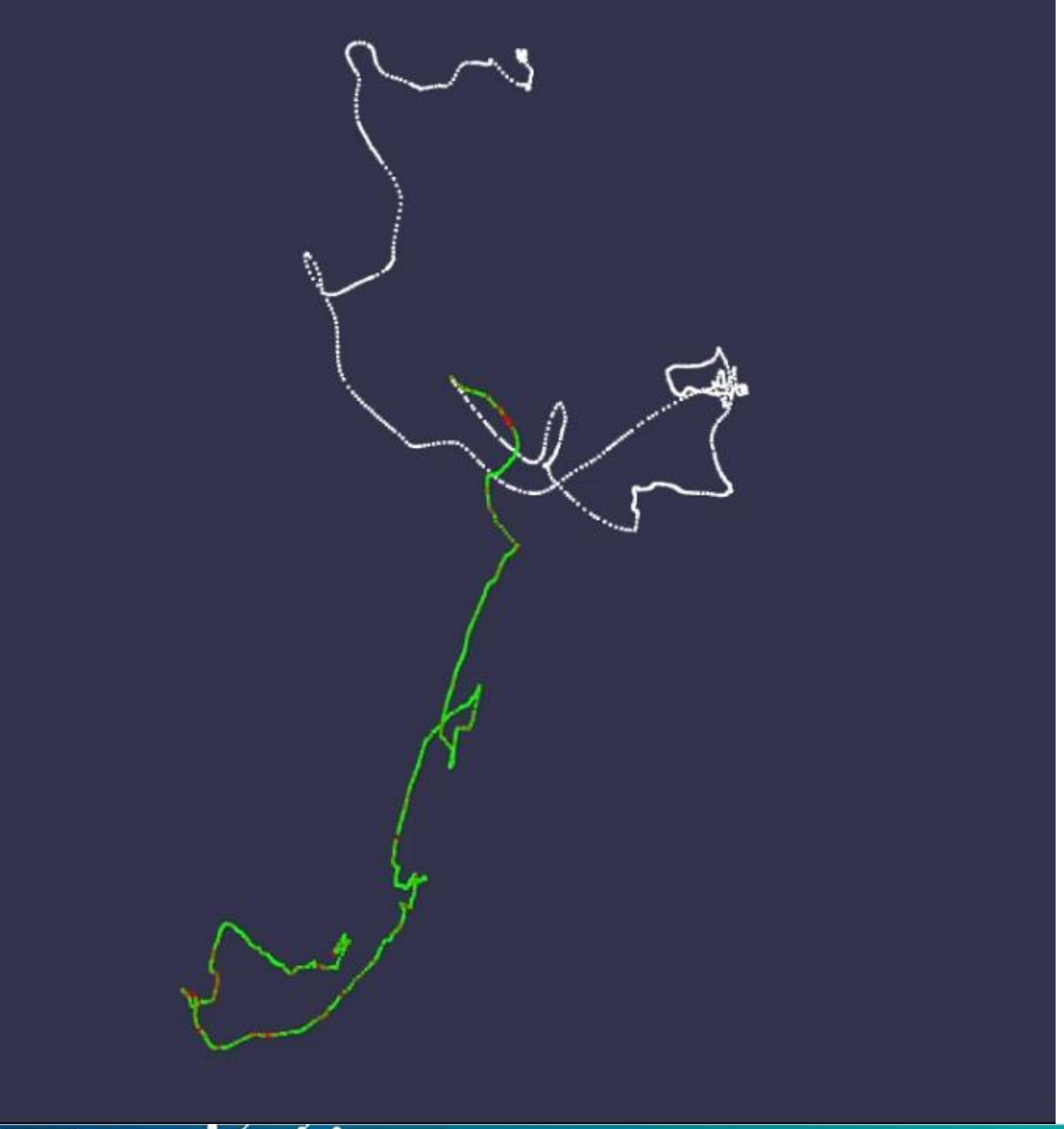

Compared to GT

- Absolute Trajectory Error: **11.68 mm**
- Relative Position Error: **1.1 mm**
- Relative Rotation Error: **2.69°**

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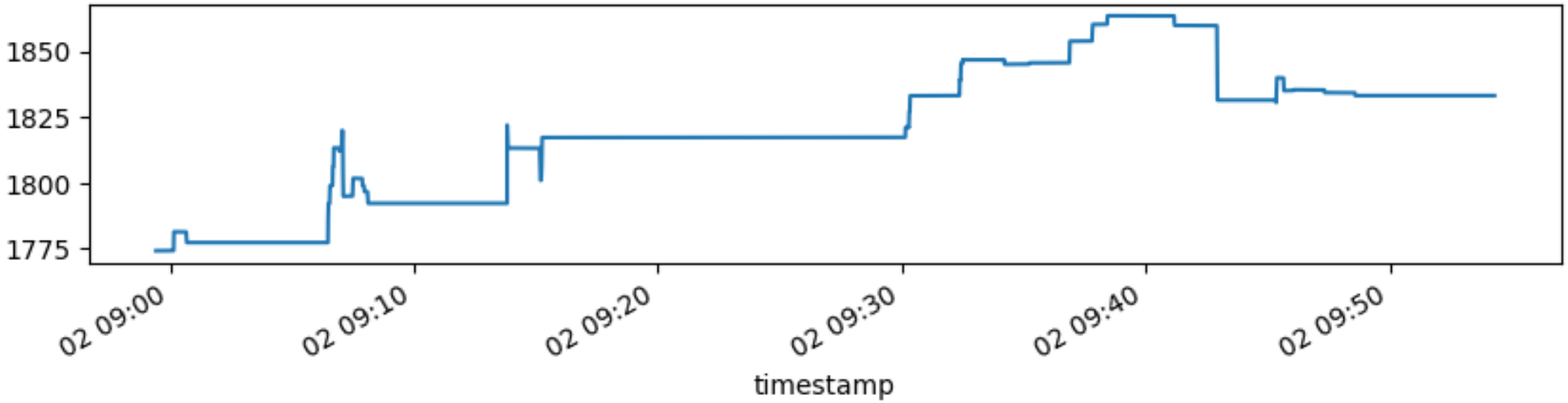
Pose Estimation - Issues

Camera Position (X coord) of a real procedure from Asensus Cloud



- Large periods with still camera, short motion periods

→ **Might not suffer this heavily from drift,** needs testing

Video-Parquet synchronization? →

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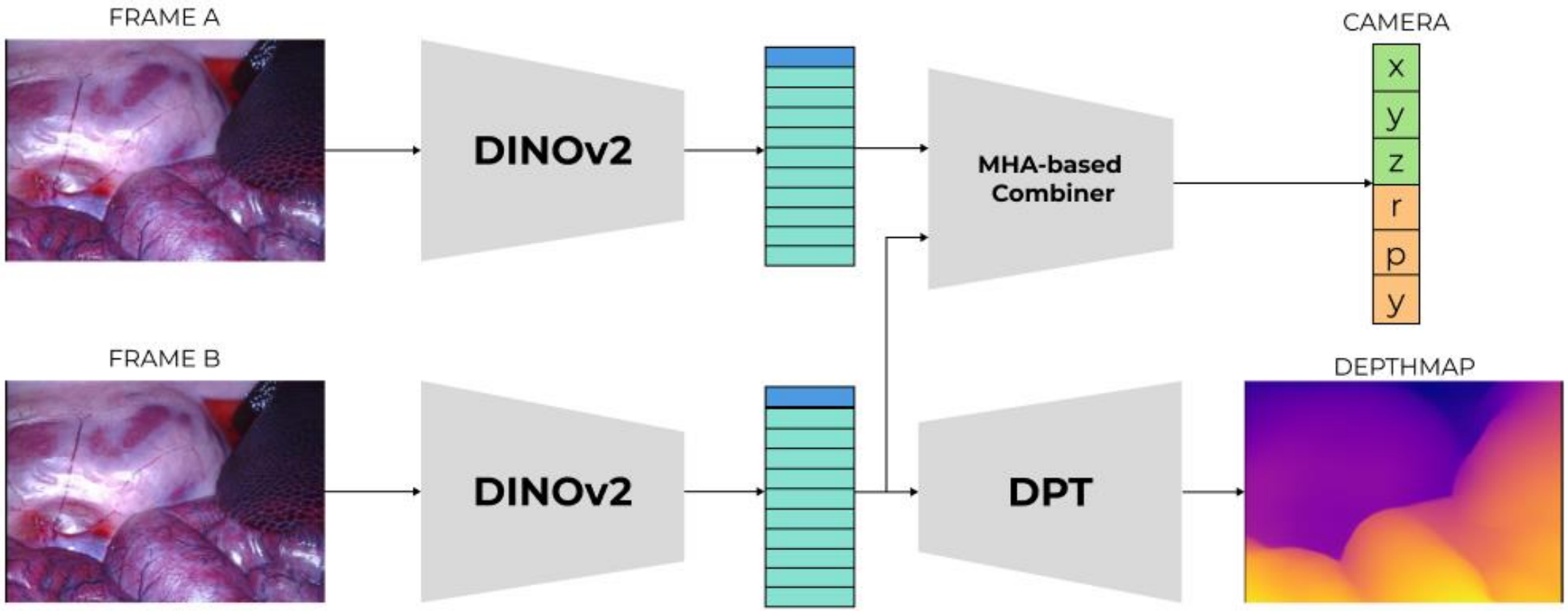
3D from Monocular Framework



Forward pass time : 1 Encoder forward pass per frame

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Work in Progress

- Depth and Pose have been trained separately

→ **Working on a joint self-supervised training** (with the Pose Network pretrained fully supervised)

- Odometry Estimation suffers from Drift
- Depth Estimation suffers from Intra-frame scale inconsistency

→ **Issues will be attenuated with joint self-supervised training**

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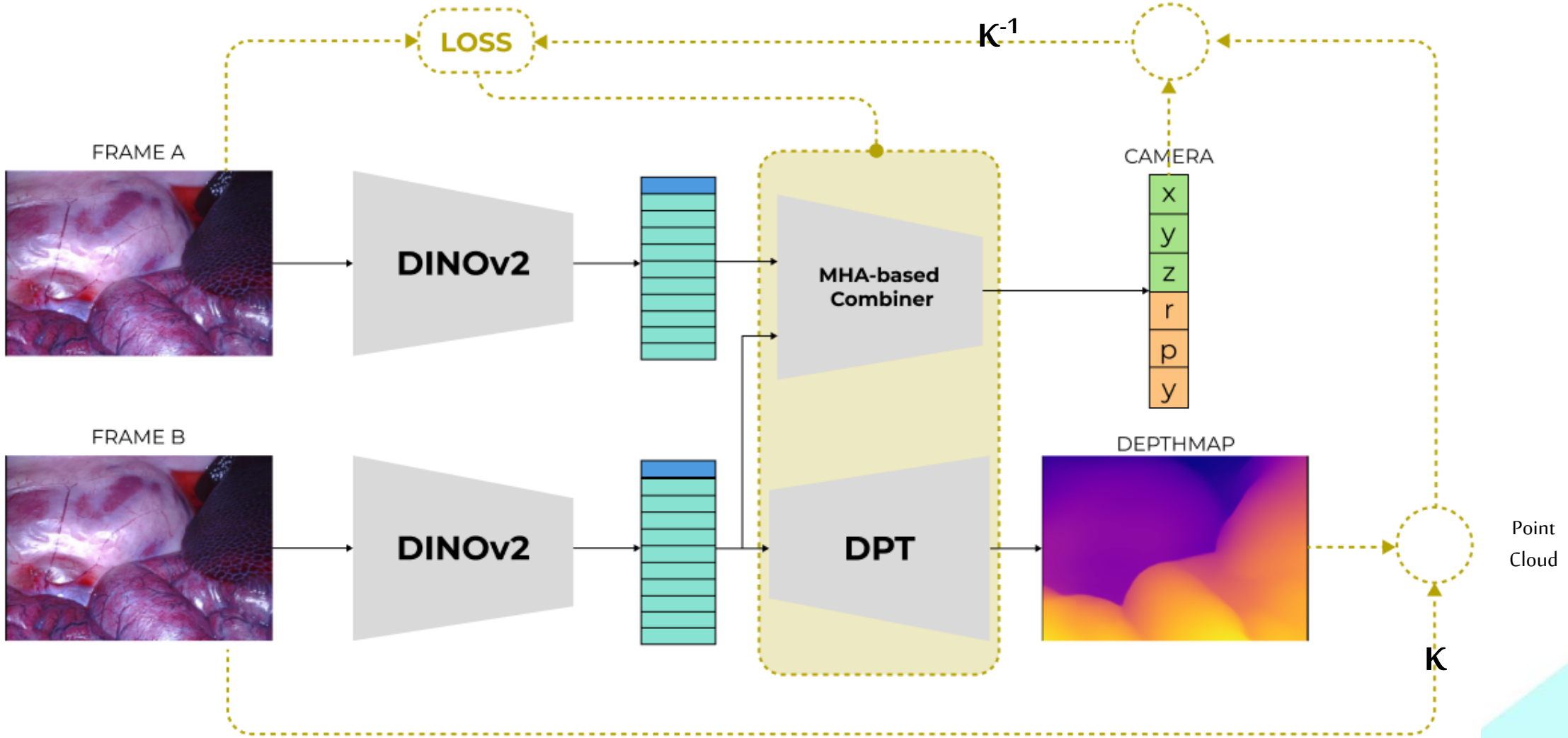
3D from Monocular Framework



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